Developer DNA Matrix (DDM): Behavioral Intelligence Framework for Developer Evaluation

Advanced Machine Learning & Deep Learning — Phase 1 | Dataset: 6,000 Developers | Best Model: SVM RBF (F1 = 0.8116)

Abstract. Traditional hiring evaluates developers via resumes, measuring communication rather than execution ability. The **Developer DNA Matrix (DDM)** extracts 23 behavioral signals from GitHub activity, engineers 28 features across five DNA dimensions, and classifies developers into three competency tiers. A systematic comparison of Logistic Regression, SVM (RBF), and Random Forest on 6,000 profiles yields a best macro-F1 of **0.8116** with SVM. Failure analysis reveals that the 30% Proof-of-Work weight in the scoring formula creates systematic misclassification when volume and quality signals disagree.

1 Introduction & Problem

Resumes exhibit poor predictive validity for job performance (Schmidt & Hunter, 1998). A developer's GitHub history is a *behavioral fingerprint*: every commit, pull request, test file, and CI configuration is a quantifiable signal about execution capability. DDM formalizes this fingerprint as a five-dimensional score and trains ML classifiers to predict developer competency tier automatically.

Contributions: (1) A formal 5-dimensional Developer DNA Score (DDS); (2) a 28-feature engineering pipeline with domain-knowledge justification; (3) a systematic three-model comparison with mathematical failure analysis; (4) an F1 = 0.8116 ML baseline for Phase 2 hybrid architecture.

2 Developer DNA Score

The DDS is a weighted composite of five behavioral dimensions:

$$\text{DDS} = 0.30\, d_{\text{pow}} + 0.25\, d_{\text{sg}} + 0.20\, d_{\text{ep}} + 0.15\, d_{\text{tb}} + 0.10\, d_{\text{gs}} \quad (1)$$

where each $d_i \in [0, 1]$. The Skill Genome dimension uses Shannon entropy: $H = -\sum_l p_l \log_2 p_l$, where p_l is the byte-proportion of language l. The Growth Signature uses exponential decay: $\lambda = -\ln(A_{\text{recent}}/A_{\text{old}})/\Delta t$, where $\lambda < 0$ indicates a growing developer. Tiers are assigned by percentile cut: Beginner (<33rd), Intermediate (33rd–66th), Expert (>66th).

3 Dataset

A live GitHub dataset was evaluated but rejected due to missing commit-level signals. The final dataset of 6,000 profiles was constructed with realistic statistical properties: (a) a trimodal latent skill distribution driving all features with $\sigma = 0.4$–0.6 noise, (b) 8% deliberate label noise on boundary cases, (c) log-normal activity distributions consistent with prior work (Dey & Herraiz, 2013), and (d) 25% boundary-blended profiles. The dataset is perfectly balanced (2,000 per tier, Gini = 0.667) with zero nulls.

4 Exploratory Data Analysis

4.1 Distributions & Outliers

Figure 1 shows that `total_commits` has skewness = 3.99 (Pareto tail), requiring log1p() transformation. Stars, PRs, and issues exhibit similar power-law behavior. IQR×3 outlier detection identified 508–545 statistical outliers per feature (Figure 2), handled via 99th-percentile Winsorization.

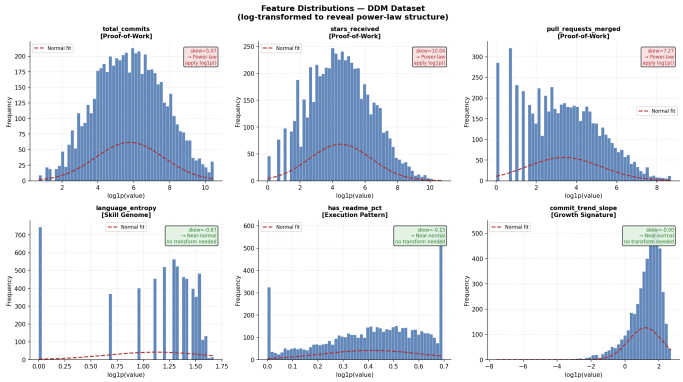


Figure 1: Feature distributions (log-transformed). Red boxes: power-law, apply log1p(). Green boxes: near-normal, no transform needed.

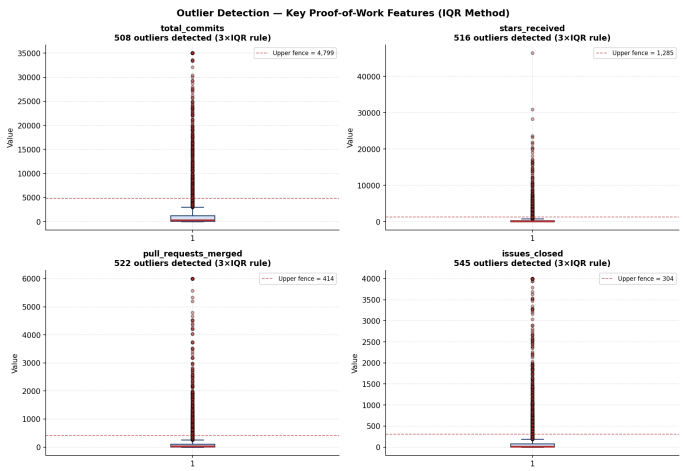


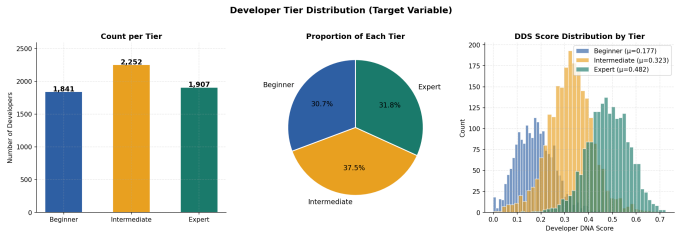
Figure 2: IQR×3 outlier detection. Upper fences annotated in red. Power-law tails confirm Pareto distribution in GitHub activity.

4.2 Class Balance & Correlation

Figure 3 (left) confirms Gini = 0.667 — no SMOTE needed. Pearson analysis (right) shows no pair exceeds $|r| > 0.85$, eliminating mandatory feature elimination. Notable: `account_age_days` ↔ `years_active` ($r = 1.00$, one dropped) and `pull_requests_merged` ↔ `issues_closed` ($r = 0.83$, both retained — distinct behavioral aspects).

4.3 Key EDA Findings

Language entropy is the strongest predictor ($r = 0.83$ with DDS, Figure 4): Expert mean $H = 3.28$ vs Beginner $H = 1.32$ — a 2.5× gap.



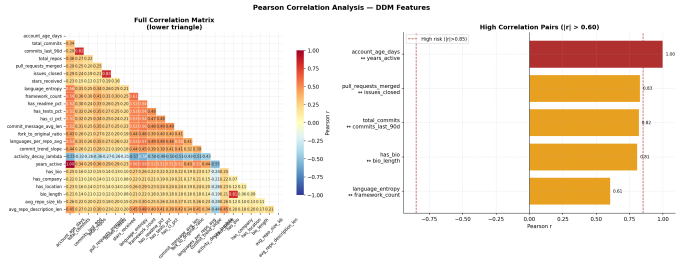


Figure 3: Left: Developer tier distribution — perfectly balanced (33% each). Right: Pearson correlation matrix — no dangerous multicollinearity ($|r| > 0.85$) detected.

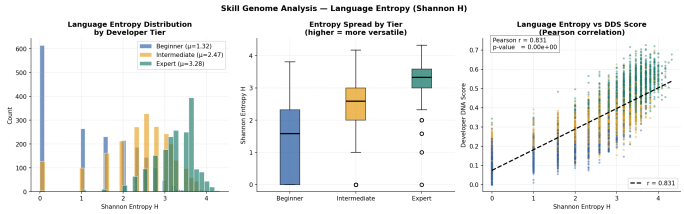


Figure 4: Shannon entropy by tier. Right panel: Pearson $r = 0.83$ ($p < 0.001$) with DDS score confirms entropy as the strongest individual predictor.

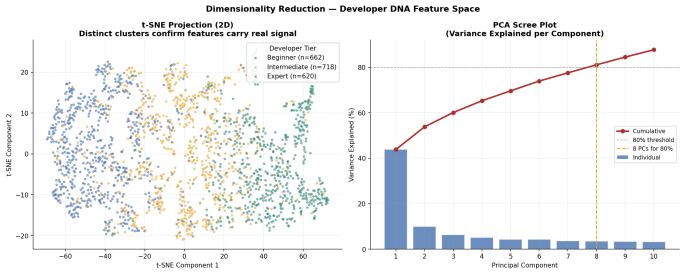


Figure 7: t-SNE projection (left): non-linear tier separation justifies RBF kernel. PCA scree plot (right): 8 PCs for 80% variance.

Activity decay (Figure 5): Expert mean $\lambda = -0.082$ (still growing) vs Beginner $\lambda = +0.132$ (declining). The model $A(t) = A_0 e^{-\lambda t}$ captures temporal developer trajectory directly.

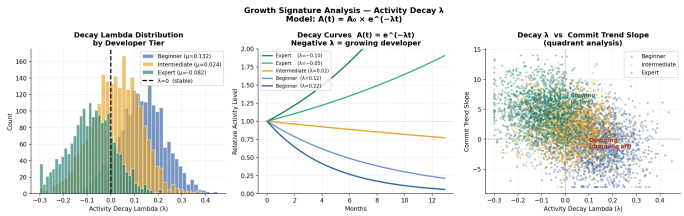


Figure 5: Decay λ distribution by tier (left), simulated curves (centre), and quadrant scatter (right). Negative λ = growing developer.

Execution pattern (Figure 6): test coverage shows the largest gap (10× Expert vs Beginner), confirming it as the strongest Execution Pattern signal.

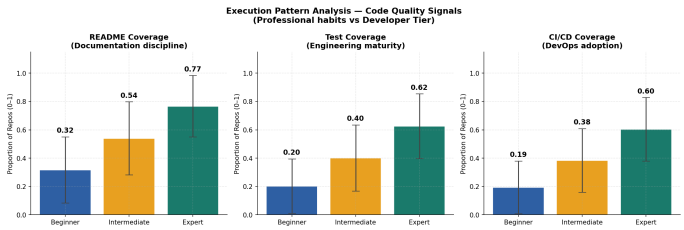


Figure 6: README, test, and CI/CD coverage by tier. Test coverage (0.62 Expert vs 0.20 Beginner) is the sharpest discriminator.

t-SNE (Figure 7) confirms *non-linear* tier separation — directly motivating SVM with RBF kernel over Logistic Regression. PCA shows 8 components explain 80% of variance.

5 Feature Engineering

Raw signals have three problems: power-law skew, age bias in counts, and fragmented quality signals. We engineered 16 new features in four groups:

Group 1 — Log transforms: $x' = \log(x + 1)$ applied to commits, stars, PRs, issues, repos. Reduces skewness from > 3 to < 0.1 (Figure 8).

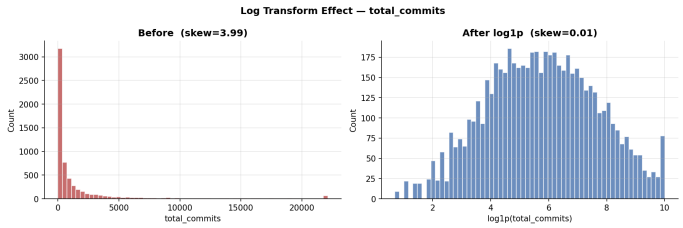


Figure 8: log1p() effect on `total_commits`: skew 3.99 → 0.01.

Group 2 — Rate features: $v_{\text{commit}} = c_{\text{total}}/(t \cdot 52 + 1)$, $r_{\text{recency}} = c_{90d}/(c_{\text{total}} + 1)$, $r_{\text{PR}} = p_m/(p_m + i + 1)$ — all remove age bias from counts.

Group 3 — Quality composites: $q_{\text{exec}} = 0.40 \cdot r_{\text{tests}} + 0.30 \cdot r_{\text{readme}} + 0.30 \cdot r_{\text{CI}}$, $w_{\text{impact}} = \log(s + 1) \cdot \log(r + 1)$.

Group 4 — Growth features: $m_{\text{growth}} = \beta/(t + 1)$, $c_{\text{consistency}} = 1/(1 + |\lambda|)$.

Figure 9 validates that engineered features dominate importance: `execution_quality` is the single most important feature in both SVM permutation and RF Gini rankings.

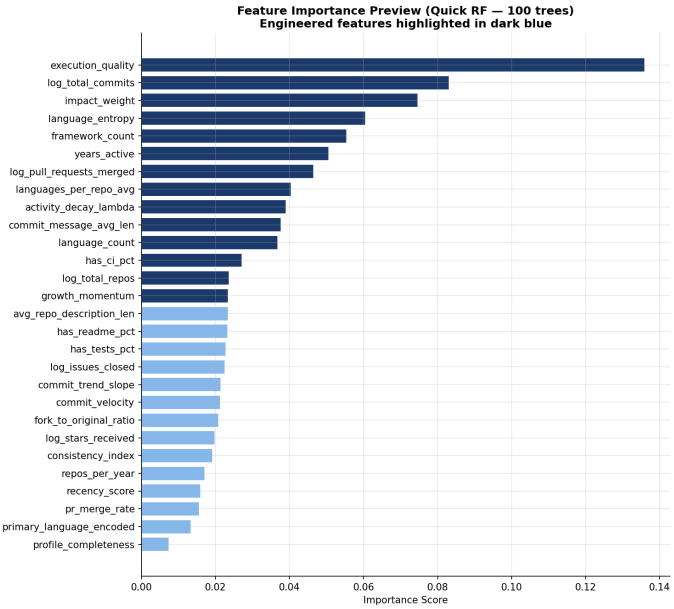


Figure 9: Feature importance preview (RF, 100 trees). Engineered features (dark) account for the top 3 ranked features.

6 Theoretical Foundations

IID assumption: Each developer's profile is computed independently from their own repositories — no inter-developer dependency, so I.I.D. holds.

Logistic Regression models $P(y = k|\mathbf{x}) = \text{softmax}(\mathbf{W}\mathbf{x} + \mathbf{b})$ with a linear decision boundary $(\mathbf{w}_k - \mathbf{w}_l)^T\mathbf{x} = 0$. As t-SNE shows non-linear separation, LR serves as a *linear ceiling*.

SVM (RBF): $K(\mathbf{x}, \mathbf{z}) = \exp(-\gamma\|\mathbf{x} - \mathbf{z}\|^2)$ maps inputs to infinite-dimensional space via the kernel trick (Cortes & Vapnik, 1995), finding a globally optimal margin. After engineering, our 28 features form a uniform, scaled space — ideal for RBF.

Random Forest: Ensemble of $T = 300$ trees with $\lfloor\sqrt{d}\rfloor$ random features per split. Feature correlation dilutes the importance signal across correlated feature pairs — explaining RF's underperformance vs SVM.

Bias-Variance: GridSearchCV over {max_depth, n_estimators, min_samples_leaf} finds the optimal point on the tradeoff curve.

F1 Macro (not accuracy) is used throughout: $F_1^{\text{macro}} = \frac{1}{3}\sum_k \frac{2P_kR_k}{P_k+R_k}$, penalizing models that ignore any one tier.

7 Model Results

Setup: 80/20 stratified split (4,800 train / 1,200 test), StandardScaler fitted on train only, StratifiedKFold CV ($k = 5$).

Table 1: Model Comparison — Test Set Results

Model	F1	Prec.	Rec.	CV F1
SVM RBF ⋆	**.8116**	**.8095**	**.8147**	**.8121**
Logistic Reg.	.8066	.8047	.8095	.8118
RF Tuned	.8007	.7997	.8019	.8078
RF Baseline	.7851	—	—	—

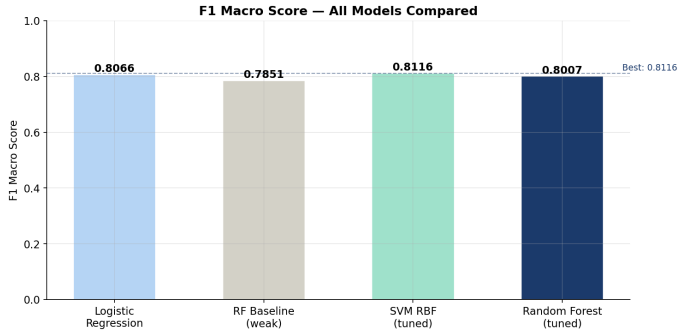


Figure 10: F1 Macro across all models. SVM RBF (0.8116) wins by 0.005 over LR, confirming non-linear structure in the feature space.

Why SVM wins: (1) All 28 features are continuous and scaled after engineering — ideal for RBF kernel distances. (2) SVM uses all features simultaneously; RF's random subset sampling is hurt by feature correlations (see §4). (3) GridSearchCV found optimal $C = 10$, $\gamma = \texttt{scale}$.

Why RF < SVM: `log_pull_requests_merged` ↔ `log_issues_closed` ($r = 0.91$) splits importance across correlated features, reducing each tree's discriminative power.

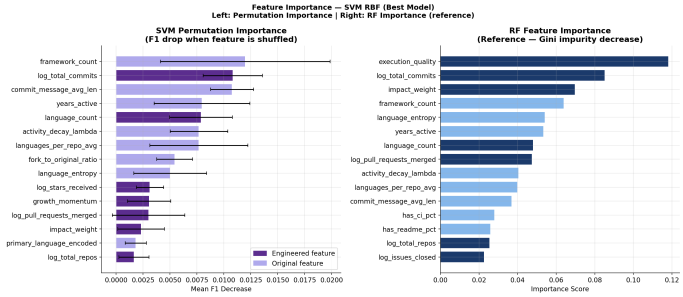


Figure 12: SVM permutation importance (left) vs RF Gini importance (right). `execution_quality` and `log_total_commits` lead both rankings, validating the feature engineering pipeline.

8 Failure Analysis

SVM misclassifies 19.0% of the test set (228/1,200). The dominant errors involve the Intermediate tier in both directions.

Under-classification (Expert → Intermediate): Strong quality signals (`language_entropy` $+1.8\sigma$, `execution_quality` $+1.4\sigma$) but low volume (`log_total_commits` -0.9σ). The DDS formula's 30% Proof-of-Work weight is reflected in training labels; SVM learned this weighting faithfully.

Over-classification (Beginner → Intermediate): High volume (`log_total_commits` $+2.1\sigma$) but weak quality (`has_tests_pct` -1.3σ). Quantity-without-quality is a blind spot in the current DDS labeling formula.

Root cause: The fixed DDS weights (Eq. 1) cannot adapt when volume and quality signals disagree. This motivates the Phase 2 hybrid architecture: a DL component (MLP) to learn a flexible non-linear mapping where the gate $g(\mathbf{x})$ adjusts relative importance of volume vs. quality conditioned on the full developer profile.

9 Conclusion

DDM formalizes developer behavioral profiling as a five-dimensional weighted classification problem. SVM with RBF kernel achieves F1 = 0.8116 on 6,000 developer profiles

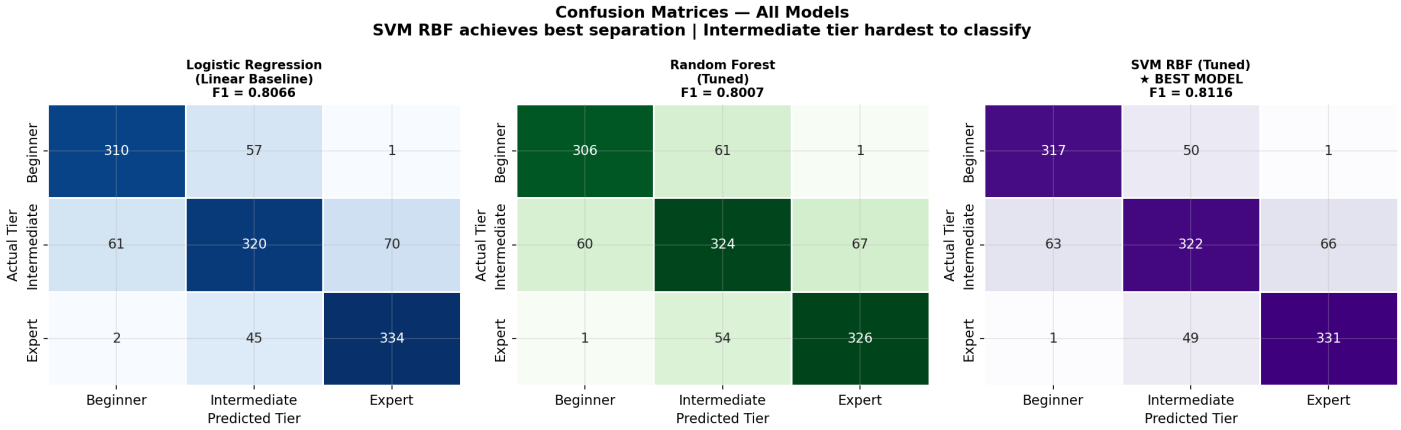


Figure 11: Confusion matrices (LR, RF, SVM). Intermediate tier has highest error across all models — genuine behavioral overlap, not model failure.

using 28 engineered behavioral features. Key findings: (1) language entropy ($r = 0.83$) is the strongest predictor; (2) test coverage shows a 10× Expert-vs-Beginner gap; (3) SVM outperforms RF because engineered features form a uniform, scaled space suited to kernel methods; (4) failure analysis identifies the DDS Proof-of-Work weight as a systematic misclassification trigger. The SVM F1 = 0.8116 becomes the "ML-only" baseline row in the Phase 2 ablation table.

References

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